

Section I - (5 marks). Attempt Questions 1-5.

Use the multiple-choice answer sheet on the previous page.

- 1 Which one of the following vectors is parallel to vector $\begin{bmatrix} -3 \\ 7 \end{bmatrix}$ and has magnitude of $2\sqrt{58}$?
- (A) $\begin{bmatrix} -24 \\ 28 \end{bmatrix}$ (B) $\frac{-3}{2}\mathbf{i} + \frac{7}{2}\mathbf{j}$ (C) $\begin{bmatrix} 3 \\ -7 \end{bmatrix}$ (D) $-6\mathbf{i} + 14\mathbf{j}$
- 2 The vectors $\mathbf{a} = 3\mathbf{i} - x\mathbf{j}$ and $\mathbf{b} = 2\mathbf{i} + 5\mathbf{j}$ are perpendicular. What is the value of x ?
- (A) $\frac{-15}{2}$ (B) $\frac{-6}{5}$ (C) $\frac{6}{5}$ (D) $\frac{15}{2}$
- 3 The angle between the vectors $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - \mathbf{j}$ is closest to:
- (A) 89° (B) 91° (C) 75° (D) 105°
- 4 In the parallelogram $ABCD$, the point of intersection of their diagonals is O . $\overrightarrow{AB} = \mathbf{a}$, $\overrightarrow{BC} = \mathbf{b}$.
The vector $\overrightarrow{OD}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ is equal to:
- (A) $\frac{1}{2}(\mathbf{a} + \mathbf{b})$ (B) $\frac{1}{2}(\mathbf{a} - \mathbf{b})$ (C) $\frac{1}{2}\mathbf{b} + \mathbf{a}$ (D) $\frac{1}{2}(\mathbf{b} - \mathbf{a})$
- 5 The vector projection of $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ onto $\mathbf{b} = -3\mathbf{i} - \mathbf{j}$ is equal to:
- (A) $\frac{3}{\sqrt{10}}\mathbf{i} + \frac{1}{\sqrt{10}}\mathbf{j}$ (B) $\frac{-3}{10}\mathbf{i} - \frac{1}{10}\mathbf{j}$ (C) $\frac{3}{10}\mathbf{i} + \frac{1}{10}\mathbf{j}$ (D) $\frac{-3}{\sqrt{10}}\mathbf{i} - \frac{1}{\sqrt{10}}\mathbf{j}$

End of Section I

Section II - (28 marks). Attempt Questions 6-10. Answer each question in the space provided. Extra writing space is provided at the back. If you use this space, clearly indicate which question you are answering.

Question 6 (11 marks)

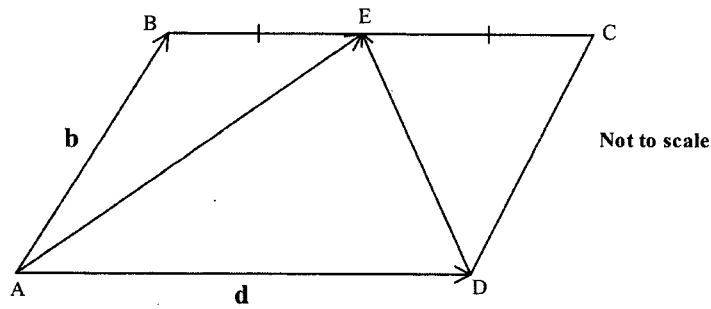
Marks

- 2

- 3**

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- c) $ABCD$ is a parallelogram in which $\overrightarrow{AB} = \mathbf{b}$, $\overrightarrow{AD} = \mathbf{d}$ and E is the midpoint of BC .



- i) Express $\overrightarrow{AE}$ and $\overrightarrow{DE}$ in terms of $\mathbf{b}$ and $\mathbf{d}$.

2

- ii) If F is a point on $\overrightarrow{DE}$ and $\overrightarrow{DF} = \frac{2}{3}\overrightarrow{DE}$, express $\overrightarrow{DF}$ in terms of $\mathbf{b}$ and $\mathbf{d}$.

1

- iii) Find $\overrightarrow{AF}$ in terms of $\mathbf{b}$ and $\mathbf{d}$ and hence show that F lies on $\overrightarrow{AC}$.

3

Lined area for writing answers.

End of question 6

Question 7 (3 marks)

Prove by mathematical induction that for all positive integers $n \geq 1$

3

$$1 \times 3 + 2 \times 4 + 3 \times 5 + \cdots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$$

End of question 7

Question 8 (3 marks)

Prove by mathematical induction that for all integers $n \geq 0$ $3^{2n} + 7$ is divisible by 8.

3

End of Question 8

Question 9 (7 marks)

The points A, B and C have position vectors $\overrightarrow{OA} = -2\mathbf{i} - 3\mathbf{j}$, $\overrightarrow{OB} = 2\mathbf{i} + 3\mathbf{j}$ and $\overrightarrow{OC} = 8\mathbf{i} - \mathbf{j}$.

- a) Find the vectors $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{AC}$.

3

- b) Show that $\triangle ABC$ is a right-angled isosceles triangle with a right angle at B .

2

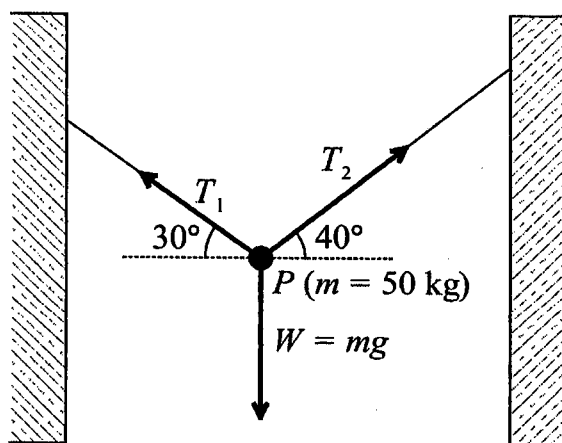
- c) Find the position vector of the point D such that $ABCD$ forms a square.

2

End of Question 9

Question 10 (4 marks)

The diagram shows an object, P , of mass 50 kg being suspended by two light inextensible strings of different lengths. The strings are at different angles to the horizontal. One is at 30° and the other is at 40° . The tension in these strings are T_1 and T_2 respectively. Take $g = 9.8\text{ m/s}^2$.



- a) Show that if the object is stationary, $T_1 = \frac{\cos(40^\circ)}{\cos(30^\circ)} T_2$.

1

b) Hence find the tension in each string if the object is stationary, correct to the nearest newton.

3

End of Paper

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

X1- 2019 December Task

Section I - Multiple Choice (5 marks) Circle the correct answer.

- 1 Which one of the following vectors is parallel to vector $\mathbf{v} = \begin{bmatrix} -3 \\ 7 \end{bmatrix}$ and has magnitude of $2\sqrt{58}$?

(A) $\begin{bmatrix} -24 \\ 28 \end{bmatrix}$ (B) $\frac{-3}{2}\mathbf{i} + \frac{7}{2}\mathbf{j}$ (C) $\begin{bmatrix} 3 \\ -7 \end{bmatrix}$ (D) $-6\mathbf{i} + 14\mathbf{j}$

(B) & (D) have same as $\mathbf{v}$, but (D) has magnitude $\sqrt{(-6)^2 + 14^2} = 2\sqrt{58}$

- 2 The vectors $\mathbf{a} = 3\mathbf{i} - x\mathbf{j}$ and $\mathbf{b} = 2\mathbf{i} + 5\mathbf{j}$ are perpendicular. What is the value of x ?

(A) $\frac{-15}{2}$ (B) $\frac{-6}{5}$ (C) $\frac{6}{5}$ (D) $\frac{15}{2}$

$\mathbf{a} \perp \mathbf{b}$ if $\mathbf{a} \cdot \mathbf{b} = 0 \therefore 3 \times 2 + (-x) \times 5 = 0 \therefore x = 6/5$

- 3 The angle between the vectors $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - \mathbf{j}$ is closest to:

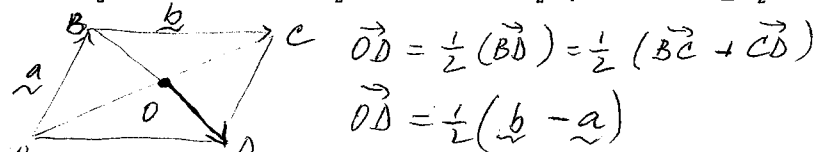
(A) 89° (B) 91° (C) 75° (D) 105°

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} = \frac{2 \times 3 + 3 \times -1}{\sqrt{13} \times \sqrt{10}} \therefore \theta = 74.74^\circ$$

- 4 In the parallelogram ABCD, the point of intersection of their diagonals is O. $\overrightarrow{AB} = \mathbf{a}$, $\overrightarrow{BC} = \mathbf{b}$.

The vector $\overrightarrow{OD}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ is equal to:

(A) $\frac{1}{2}(\mathbf{a} + \mathbf{b})$ (B) $\frac{1}{2}(\mathbf{a} - \mathbf{b})$ (C) $\frac{1}{2}(\mathbf{b}) + \mathbf{a}$ (D) $\frac{1}{2}(\mathbf{b} - \mathbf{a})$



- 5 The vector projection of $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ onto $\mathbf{b} = -3\mathbf{i} - \mathbf{j}$ is equal to:

(A) $\frac{1}{\sqrt{10}}(3\mathbf{i} + \mathbf{j})$ (B) $\frac{1}{10}(-3\mathbf{i} - \mathbf{j})$ (C) $\frac{1}{10}(3\mathbf{i} + \mathbf{j})$ (D) $\frac{1}{\sqrt{10}}(-3\mathbf{i} - \mathbf{j})$

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}} \right) \mathbf{b} = \frac{(2\mathbf{i} - 5\mathbf{j}) \cdot (-3\mathbf{i} - \mathbf{j})}{(-3\mathbf{i} - \mathbf{j}) \cdot (-3\mathbf{i} - \mathbf{j})} (-3\mathbf{i} - \mathbf{j})$$

$$= \frac{-6 + 5}{10} (-3\mathbf{i} - \mathbf{j}) = -\frac{1}{10}(-3\mathbf{i} - \mathbf{j}) = \frac{1}{10}(3\mathbf{i} + \mathbf{j})$$

End of Section I

Section II - (28 marks). Attempt Questions 6-10. Answer each question in the space provided. Extra writing space is provided at the back. If you use this space, clearly indicate which question you are answering.

Question 6 (11 marks)

Marks

- a) The point C(-7,13) has a position vector of $\mathbf{c}$. Find the vector $\overrightarrow{OA}$ if $\overrightarrow{AB}$ is equal to $\mathbf{c}$ and B is the point (12, -2).

2

Marking criteria	Marks
$\mathbf{c} = -7\mathbf{i} + 13\mathbf{j} = \overrightarrow{OC}$ $\overrightarrow{OB} = 12\mathbf{i} - 2\mathbf{j}$	
since $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$, let $\overrightarrow{OA} = x\mathbf{i} + y\mathbf{j}$	correct solns 2
$\therefore -7\mathbf{i} + 13\mathbf{j} = (12\mathbf{i} - 2\mathbf{j}) - (x\mathbf{i} + y\mathbf{j})$ ①	Establishes $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$
$\therefore -7 = 12 - x$ & $13 = -2 - y$	into equation ①
$x = 19$ $y = -15$	Finds coord.
$\therefore \overrightarrow{OA} = 19\mathbf{i} - 15\mathbf{j}$ or $\begin{bmatrix} 19 \\ -15 \end{bmatrix}$ ①	of A correctly ①

- or $\overrightarrow{OA}$ is a position vector of (19, -15)
 b) Find a vector $\mathbf{a}$ that is perpendicular to vector $\mathbf{b} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$ and has a magnitude of 10.

3

Marking criteria	Marks
let $\mathbf{a} = \begin{bmatrix} x \\ y \end{bmatrix}$ and given $\mathbf{b} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$	
since $\mathbf{a} \perp \mathbf{b}$	correct solns 3
$\therefore$ (1) $\mathbf{a} \cdot \mathbf{b} = 0$	establishes ②
$\therefore 3x - 4y = 0$	$\mathbf{a} \cdot \mathbf{b} = 0$ and $ \mathbf{a} = 10$ into
and since $ \mathbf{a} = 10$	simult. eqns
$\therefore$ (2) $\sqrt{x^2 + y^2} = 10$	solves the eqns

Now solve simultaneously

(1) $x = \frac{4}{3}y$ finds one of the solns. ①

$\therefore$ (2) $\frac{16}{9}y^2 + y^2 = 10^2$

$y^2 = 36 \therefore y = \pm 6$

if $y = 6$

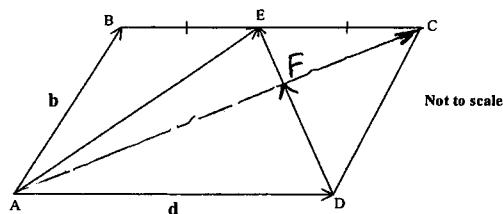
$\therefore x = 8$

OR if $y = -6$

$x = -8$

$\therefore \mathbf{a} = 8\mathbf{i} + 6\mathbf{j}$ $\therefore \mathbf{a} = -8\mathbf{i} - 6\mathbf{j}$ ①

- c) $ABCD$ is a parallelogram in which $\vec{AB} = \mathbf{b}$, $\vec{AD} = \mathbf{d}$ and E is the midpoint of BC .



- i) Express $\vec{AE}$ and $\vec{DE}$ in terms of $\mathbf{b}$ and $\mathbf{d}$.

$\vec{AE} = \vec{AB} + \vec{BE} = \mathbf{b} + \frac{1}{2}\mathbf{d}$ ①	Marking criteria	2
$\vec{DE} = \vec{DC} + \vec{CE} = \mathbf{b} - \frac{1}{2}\mathbf{d}$ ①	correct solutions.	2
	finds $\vec{AE}$ or $\vec{DE}$ correctly ①	

- ii) If F is a point on $\vec{DE}$ and $\vec{DF} = \frac{2}{3}\vec{DE}$, express $\vec{DF}$ in terms of $\mathbf{b}$ and $\mathbf{d}$.

since $\vec{DF} = \frac{2}{3}\vec{DE}$ (given)	Marking criteria	Marks
and $\vec{DE} = \mathbf{b} - \frac{1}{2}\mathbf{d}$ (from (i))	correct solution	①
$\therefore \vec{DF} = \frac{2}{3}\left(\mathbf{b} - \frac{1}{2}\mathbf{d}\right) = \frac{2}{3}\mathbf{b} - \frac{1}{3}\mathbf{d}$		

- iii) Find $\vec{AF}$ in terms of $\mathbf{b}$ and $\mathbf{d}$ and hence show that F lies on $\vec{AC}$.

$\vec{AF} = \vec{AD} + \vec{DF} = \mathbf{d} + \left(\frac{2}{3}\mathbf{b} - \frac{1}{3}\mathbf{d}\right)$		
$\therefore \vec{AF} = \frac{2}{3}\mathbf{d} + \frac{2}{3}\mathbf{b}$ ①		
Now $\vec{AC} = \vec{AD} + \vec{DC} = \mathbf{d} + \mathbf{b}$		
so to show that F lies on $\vec{AC}$		
is to show that		
$\vec{AF} = k \times \vec{AC}$ for $0 < k < 1$		
or $\vec{FC} = \lambda \times \vec{AC}$ for $0 < \lambda < 1$		

$$\text{Now } \vec{FC} = \vec{FD} + \vec{DC} = -\vec{DF} + \vec{DC}$$

$$\therefore \vec{FC} = -\left(\frac{2}{3}\mathbf{b} - \frac{1}{3}\mathbf{d}\right) + \mathbf{b}$$

from (ii)

$$\therefore \vec{FC} = \frac{1}{3}\mathbf{b} + \frac{1}{3}\mathbf{d} = \frac{1}{3}(\mathbf{b} + \mathbf{d})$$

$$\therefore \vec{FC} = \frac{1}{3} \times \vec{AC} \quad \text{AC}$$

$\therefore F$ lies on $\vec{AC}$

DB similarly $\vec{AF} = \vec{AD} + \vec{DF}$
 $= \frac{2}{3}\mathbf{d} + \frac{2}{3}\mathbf{b}$ (shown above)

$$\therefore \vec{AF} = \frac{2}{3}(\mathbf{d} + \mathbf{b})$$

$$\therefore \vec{AF} = \frac{2}{3}\vec{AC} \quad \therefore F \text{ lies on } \vec{AC}$$

Marking criteria	Marks
• correct solution	3
• finds $\vec{AF}$ and $\vec{AC}$ in terms of $\mathbf{b}$ and $\mathbf{d}$	2
• finds $\vec{AF}$ in terms of $\mathbf{b}$ and $\mathbf{d}$	①
• finds $\vec{AC}$ in terms of $\mathbf{b}$ and $\mathbf{d}$	①

End of question 6

Question 7 (3 marks)

Prove by mathematical induction that for all positive integers $n \geq 1$

3

$$1 \times 3 + 2 \times 4 + 3 \times 5 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$$

STEP 1: prove identity true for $n=1$

$$\text{LHS} = 1 \times (1+2) = 3$$

$$\text{RHS} = \frac{1(2)(9)}{6} = 3 \quad \therefore \text{identity is true for } n=1.$$

STEP 2: Now assume that identity is true for some integer $k \geq 1$.

$$\text{i.e. } 1 \times 3 + 2 \times 4 + \dots + k(k+2) = \frac{k(k+1)(2k+7)}{6}$$

STEP 3: Now prove the identity is true for $n=k+1$.

$$\text{i.e. } 1 \times 3 + 2 \times 4 + \dots + (k+1)(k+3) = \frac{(k+1)(k+2)(2k+9)}{6}$$

Proof:

$$\text{LHS} = 1 \times 3 + \dots + k(k+2) + (k+1)(k+3) =$$

$$= \frac{k(k+1)(2k+7)}{6} + (k+1)(k+3)$$

$$= \frac{k(k+1)(2k+7) + 6(k+1)(k+3)}{6}$$

$$= \frac{(k+1)[k(2k+7) + 6(k+3)]}{6}$$

$$= \frac{(k+1)(2k^2 + 13k + 18)}{6} = \frac{(k+1)(2k+9)(k+2)}{6}$$

$$= \text{RHS} \therefore \text{identity is proven true}$$

$$\text{for } n=k+1, \text{ if it's true for } n=k$$

$$\text{and since it's proven true for } n=1$$

$$\therefore \text{by mathematical induction}$$

$$\text{it's true for all integers } n \geq 1.$$

Question 8 (3 marks)

Prove by mathematical induction that for all integers $n \geq 0$

$3^{2n} + 7$ is divisible by 8.

3

STEP 1: Prove $3^{2n} + 7$ is divisible by 8

for $n=0$

$$\therefore 3^{2(0)} + 7 = 1 + 7 = 8 \text{ which is divisible by 8} \therefore 3^{2n} + 7 \text{ is div. by 8 for } n=0$$

STEP 2: Assume $3^{2n} + 7$ is div. by 8 for some integer $k \geq 0$.

$$\text{i.e. } 3^{2k} + 7 = 8m \text{ (m - integer)}$$

STEP 3: Now prove $3^{2n} + 7$ is div. by 8 for

$$n=k+1 \text{ i.e. } 3^{2(k+1)} + 7 = 8w \text{ (w - integer)}$$

$$\text{Proof: } 3^{2(k+1)} + 7 = 3^{2k} \times 3^2 + 7$$

$$\text{but from assumption } 3^{2k} = 8m - 7$$

$$\therefore 3^{2k} \times 3^2 + 7 = 3^2(8m - 7) + 7$$

$$= 9 \times 8m - 9 \times 7 + 7 = 9 \times 8m - 8 \times 7$$

$$= 8(9m - 7) = 8w \text{ (w - integer)}$$

since m is integer

$$\therefore 3^{2n} + 7 \text{ is div. by 8 for } n=k+1$$

$$\text{if it is div. by 8 for } n=k$$

$$\text{and since } 3^{2n} + 7 \text{ is div. by 8}$$

$$\text{for } n=1 \text{ (proven) } \therefore \text{by mathematical}$$

$$\text{induction } 3^{2n} + 7 \text{ is divisible by 8}$$

$$\text{for all integers } n \geq 0$$

Marking criteria - same 4 key parts as in Q.7

Criteria	Marks
• All 4 parts correct	3
• Any 3 of the 4 parts correct	2
• Any 2 of the 4 parts correct	1

End of Question 8

Question 9 (7 marks)

The points A, B and C have position vectors $\vec{OA} = -2\mathbf{i} - 3\mathbf{j}$, $\vec{OB} = 2\mathbf{i} + 3\mathbf{j}$ and $\vec{OC} = 8\mathbf{i} - \mathbf{j}$.

- a) Find the vectors $\vec{AB}$, $\vec{BC}$ and $\vec{AC}$. Criteria 3
- b) Show that $\triangle ABC$ is a right-angled isosceles triangle with a right angle at B. 2

$\vec{AB} = \vec{OB} - \vec{OA} = 4\mathbf{i} + 6\mathbf{j}$	Criteria 1 mark for each correct solution
$\vec{BC} = \vec{OC} - \vec{OB} = 6\mathbf{i} - 4\mathbf{j}$	
$\vec{AC} = \vec{OC} - \vec{OA} = 10\mathbf{i} + 2\mathbf{j}$	

show first $\vec{AB} \perp \vec{BC}$ which means

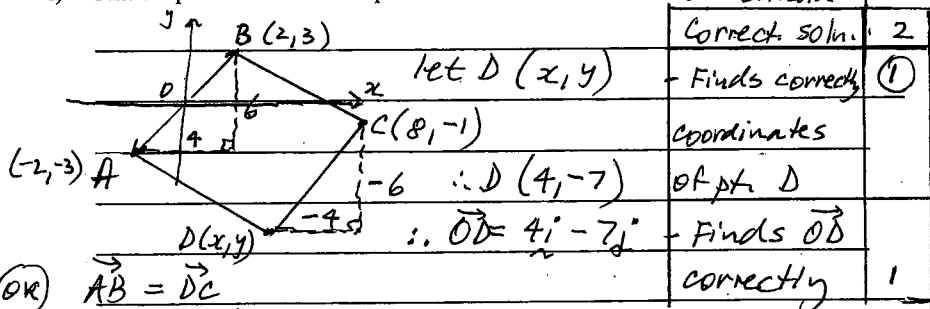
$$\vec{AB} \cdot \vec{BC} = 0$$

$$\therefore (4\mathbf{i} + 6\mathbf{j}) \cdot (6\mathbf{i} - 4\mathbf{j}) = 4 \times 6 + 6 \times -4 = 0$$

$$\therefore \vec{AB} \perp \vec{BC} \therefore \triangle ABC \text{ is right angled}$$

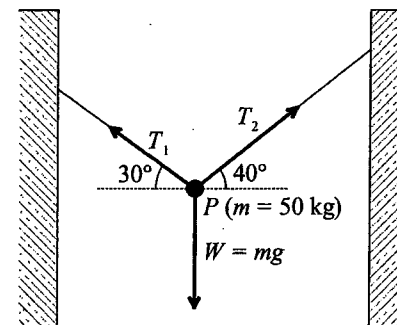
Criteria	Marks
Now show $ \vec{AB} = \vec{BC} $	1
$ \vec{AB} = \sqrt{4^2 + 6^2} = \sqrt{52}$	(using dot product or otherwise)
$ \vec{BC} = \sqrt{6^2 + (-4)^2} = \sqrt{52}$	
$\therefore \triangle ABC$ is isosceles	1

- c) Find the position vector of the point D such that ABCD forms a square. Criteria Marks 2



Question 10 (4 marks)

The diagram shows an object, P, of mass 50 kg being suspended by two light inextensible strings of different lengths. The strings are at different angles to the horizontal. One is at 30° and the other is at 40° . The tension in these strings are T_1 and T_2 respectively. Take $g = 9.8 \text{ m/s}^2$.



- a) Show that $T_1 = \frac{\cos(40^\circ)}{\cos(30^\circ)} T_2$.

let $\vec{T}_1 = x_1\mathbf{i} + y_1\mathbf{j} = T_1(\cos 30^\circ\mathbf{i} + \sin 30^\circ\mathbf{j})$

and $\vec{T}_2 = x_2\mathbf{i} + y_2\mathbf{j} = T_2(\cos 40^\circ\mathbf{i} + \sin 40^\circ\mathbf{j})$

where $|\vec{T}_1| = T_1$ and $|\vec{T}_2| = T_2$

Now since object is stationary

$$x_1 + x_2 = 0$$

$$\therefore -T_1 \cos 30^\circ + T_2 \cos 40^\circ = 0$$

$$\therefore T_1 \cos 30^\circ = T_2 \cos 40^\circ \therefore T_1 = \frac{\cos 40^\circ}{\cos 30^\circ} T_2$$

$\therefore$ shown

- b) Hence find the tension in each string if the object is stationary, correct to the nearest newton.

3

$$\text{Now } W = mg \text{ where } g = 9.8 \text{ m/s}^2$$

$$\therefore W = 50 \times 9.8 = \underline{490 \text{ N}} \quad (1)$$

and since object is stationary

$$\therefore y_1 + y_2 = W$$

$$(2) \therefore T_1 \sin 30^\circ + T_2 \sin 40^\circ = 490$$

from (a) $T_1 = \frac{\cos 40^\circ}{\cos 30^\circ} T_2$

$$\therefore \text{sub. } T_1 \text{ into (2)} \quad (1)$$

$$\therefore (2) \frac{\cos 40^\circ}{\cos 30^\circ} T_2 \times \sin 30^\circ + T_2 \sin 40^\circ = 490$$

$$T_2 \left(\frac{\cos 40^\circ}{\cos 30^\circ} \times \sin 30^\circ + \sin 40^\circ \right) = 490$$

$$T_2 = \frac{490}{\frac{\cos 40^\circ}{\cos 30^\circ} \times \sin 30^\circ + \sin 40^\circ}$$

$$\therefore T_2 = 451.586 \dots = \underline{452 \text{ N}} \quad (\text{nearest newton})$$

$$\text{now } T_1 = \frac{\cos 40^\circ}{\cos 30^\circ} \times T_2 = \frac{\cos 40^\circ}{\cos 30^\circ} \times 451.586$$

$$\therefore T_1 = 399.451 \dots = \underline{399 \text{ N}}$$

End of Paper

if used $T_2 = 452 \text{ N} \therefore T_1 = 399.8174$

$$\therefore T_1 = \underline{400 \text{ N}}$$

Part a)

Criteria

Marks

Correct solution

1

Part b)

Correct solns.

3

- Finds correctly the value W and equates $y_1 + y_2 = W$ and correctly subst. T_1 into this equation

2

- Finds tensions T_1 and T_2 correctly and finds correctly the value of W

2

- Finds correctly the value of W

1

- Equates $y_1 + y_2 = W$ correctly

1

- Evaluates T_1 & T_2 using correct approach

1